Investigating sequential n-player duel games

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The Truel

Simon Singh's *Fermat's Enigma* discusses a cute puzzle in a digression about game theory. It goes something like this:

Susan, Mary, and Bob decide to have a three-way pistol duel, a "truel." Susan hits her target every time. Mary lands her shot $\frac{2}{3}$'s of the time. Bob only lands his shot $\frac{1}{3}$ of the time. To make up for the difference in skill, they agree to fire sequentially in ascending skill: Bob goes first, followed by Mary and then Susan if they are still standing. Turns are cyclical and the game ends when all the players but one are down; we assume no coordination occurs. We'll also assume that players want to be the last player standing and, if possible, unharmed. Players may pass their turns by firing in the air. ***What should Bob do to maximize his chances of winning the duel?***

People familiar with this problem know that Bob maximizes his chances by firing into the air. Intuitively, this arises from the fact that by shooting at Mary, he risks being instantly downed by Susan. Though his risk is reduced by shooting at Susan, in the event that he hits her, Mary gets to fire at him first. By firing in the air Bob guarantees that he goes first in the inevitable duel with either Susan or Mary. Making this rigorous is a fun exercise in basic probability and series convergence.

Higher non-cooperative duel games

What happens when there are more duelists? Does the strategy "pass your turn if you are not one of the top two players" still hold for four, five, or n-person sequential duels? We make some immediate progress with the following theorem.

Theorem. *In a sequential multi-duel with n shooters $(s_1, s_2, \ldots, s_n)$ where p_i is the probability of s_i hitting their target and $p_1 < p_2 < \cdots < p_n$, if it is s_a's turn and they choose to fire at another player, they maximize their chance of surviving till their next turn by shooting at s_b if*

$$p_b = max\Big\{p_i \mid i \in \{1, \ldots, n\}\backslash\{a\}\Big\}$$

Proof. We note that in order to win, players need to survive until they are the final player and that at each cycle of turns, they are presented with a choice that can make survival more or less likely. Then in an n-player sequential multiduel with shooters $s_1, s_2, \ldots, s_n$ where shooter s_i hits their target with probability p_i and $p_1 < p_2 < \cdots < p_n$, the probability that s_a survives till their next turn if they fire at s_b is

$$(p_a)\sum_{i=0}^{n-2}\Big(\prod_{j\in S_{ik}\backslash\{b\}}(1-p_j)\Big) + (1-p_a)\sum_{i=0}^{n-1}\Big(\prod_{j\in S_{ik}}(1-p_j)\Big)$$

where S_{ik} is the kth, i-element subset of $\{1, 2, \ldots, n\}\backslash\{a\}$ under some canonical ordering. Now, since $p_i < p_n$ for every $i \in \{1, 2, \ldots, n-1\}$, it is the case that $(1-p_i) > (1-p_n)$ by the definition of a probability. Therefore, it is the case that

$$\sum_{i=0}^{n-2}\Big(\prod_{j\in S_{ik}\backslash\{n\}}(1-p_j)\Big) > \sum_{i=0}^{n-2}\Big(\prod_{j\in S_{ik}\backslash\{i\}}(1-p_j)\Big)$$

because every occurrence of $(1-p_n)$ in the latter sum of products is replaced with an instance of the larger value $(1-p_i)$ in the former. It immediately follows then that for every a and i in $\{1, \ldots, n-1\}$

$$p(s_a \text{ survives} \mid s_a \text{ fires at } s_n) > p(s_a \text{ survives} \mid s_a \text{ fires at } s_i)$$

An identical argument with edited indices will reveal that s_n must fire at s_{n-1} in order to maximize their chances of surviving till their next turn. Thus, s_a maximizes their chance of surviving till their next turn by firing at s_b where

$$p_b = \max\Big\{p_i \mid i \in \{1, \ldots, n\}\backslash\{a\}\Big\}.$$

■

The above proof is necessary because it allows us to definitely dismiss any strategies where player i fires at someone who is weaker than the best, non-self player. This significantly narrows the scope of what *might be optimal* for player one to do in a multiduel. As things stand, players must always either fire at the best shot, or do nothing. After some calculation, we can immediately say the following:

Corollary. *The optimal strategy for s_{n-1} and s_n is to always fire at each other.*

Optimal strategy for the nth and $(n-1)^{th}$ players is fixed, but it is less clear what players 1 through $n-2$ should do to maximize their survival.

Non-cooperative duel for 4 players

Let P_1, P_2, P_3, P_4 be our duelists. They fire sequentially in ascending order and player i has a probability of $\frac{i}{4}$ of landing their shot. What strategy should should P_1 adopt to maximize their chances of winning the duel?

We have already demonstrated that every player, if they are going to fire at another player, should always fire at the best player, but there is some variety in the possible strategies when we recall that players may also pass their turn. There are a few strategies we should consider:

"Revolution"	Everyone fires at the best player.
"Patience"	Only the top two players fire at each other, everyone else passes.
"Desperation"	First player fires at best initially, everyone else adopts "Patience."

Conspicuously absent from the list of possible player strategies is the one where P_1 passes and P_2 fires. Simple calculation reveals that P_2 survives to the next round with probability 1 by not firing at Players 3 or 4. They survive with a probability less than 1 if they do fire, disqualifying this choice for P_2.

One can get an idea about each strategy's performance with a simulation. We wrote one in Python and have included a version for each strategy in Appendix A. Here are some simulated probabilities of a given player winning in one of our candidates for optimal global strategy.

Revolution Simulation Probabilities

P_1	P_2	P_3	P_4
0.32083	0.30505	0.34726	0.02686

Patience Simulation Probabilities

P_1	P_2	P_3	P_4
0.35091	0.33362	0.22094	0.09453

Desperation Simulation Probabilities

P_1	P_2	P_3	P_4
0.35474	0.33162	0.24204	0.0716

Preliminary results suggest that the Desperation strategy is the best strategy. We'll attempt to confirm this with computations.

Probabilities of Victory: "Desperation"

We first consider the likelihood that P_1 wins a duel with each of the four players, given that P_1 enters into it (we'll compute the likelihood of entering such duels later). This

isn't too hard, P_1's strategy guarantees first shot in any duel, so we may assume that P_1 goes first. Taking $P_i \text{☠} P_j$ to mean "P_i hits P_j" and $P_i \circ P_j$ to mean "P_i misses P_j," we have

$$P(P_1 \text{☠} P_2 \mid \text{they duel}) = \frac{1}{4}\sum_{n=0}^{\infty}\left(\frac{3}{4}\right)^n\left(\frac{1}{2}\right)^n = \frac{1}{4}\sum_{n=0}^{\infty}\left(\frac{3}{8}\right)^n = \frac{1}{4}\left(\frac{8}{5}\right) = \frac{2}{5}$$

$$P(P_1 \text{☠} P_3 \mid \text{they duel}) = \frac{1}{4}\sum_{n=0}^{\infty}\left(\frac{3}{4}\right)^n\left(\frac{1}{4}\right)^n = \frac{1}{4}\sum_{n=0}^{\infty}\left(\frac{3}{16}\right)^n = \frac{1}{4}\left(\frac{16}{13}\right) = \frac{4}{13}$$

$$P(P_1 \text{☠} P_4 \mid \text{they duel}) = \frac{1}{4}$$

Now we need to know the probability that P_1 enters into a duel with each player.

A potential shortcut might be to try to figure out what the probability is that each player is left alone with P_1.

Probability of dueling P_4

This is the simplest case. P_1 duels P_4 if

a. $P_4 \circ P_1$ ($\frac{3}{4}$)

b. $P_4 \circ P_3$ ($\frac{1}{4}$) and $P_4 \text{☠} P_3$ (1)

c. $P_4 \circ P_2$ ($\frac{1}{2}$) and $P_4 \text{☠} P_2$ (1)

So P_1 duels P_4 with probability $\frac{3}{32}$.

Probability of dueling P_3

Now, P_3 duels P_1 if and only if P_3 enters into, and wins, a mini-duel with P_2. This happens if

a. $P_1 \text{☠} P_4$ ($\frac{1}{4}$) or

b. $P_1 \circ P_4$ ($\frac{3}{4}$) and $P_3 \text{☠} P_4$ ($\frac{3}{4}$)

We note that in each situation, P_2 gets the first strike in the resulting duel. So we have:

$$P(P_3 \text{☠} P_2 \mid \text{they duel}) = \frac{3}{4}\sum_{n=0}^{\infty}\left(\frac{1}{2}\right)^{n+1}\left(\frac{1}{4}\right)^n = \frac{3}{8}\sum_{n=0}^{\infty}\left(\frac{1}{8}\right)^n = \frac{3}{8}\left(\frac{8}{7}\right) = \frac{3}{7}$$

So the probability that P_1 duels P_3 is

$$\frac{3}{7}\left(\frac{1}{4} + \frac{9}{16}\right) = \frac{3}{7}\left(\frac{13}{16}\right) = \frac{39}{112}$$

Probability of dueling P_2

In order for P_1 to face P_2, P_2 must enter into and win a mini duel with either P_3 or P_4. Fortunately, we've already worked out some of these details. We know that P_3 enters a mini-duel with P_2 with probability $\frac{13}{16}$ and that P_2 wins said duel with probability $1 - \frac{3}{7} = \frac{4}{7}$. P_4 ends up dueling P_2 with probability $\frac{3}{16}$ and P_2 wins with probability $\frac{1}{2}$. Thus, we have

$$P(P_1 \text{ duels } P_2) = \left(\frac{13}{16}\right)\left(\frac{4}{7}\right) + \left(\frac{3}{16}\right)\left(\frac{1}{2}\right) = \frac{125}{224}$$

Putting it all together

The probability of P_1 winning the duel with the "Desperation" strategy is

$$\left(\frac{125}{224}\right)\left(\frac{2}{5}\right) + \left(\frac{39}{112}\right)\left(\frac{4}{13}\right) + \left(\frac{3}{32}\right)\left(\frac{1}{4}\right) = \frac{317}{896} \approx 0.3537$$

Some additional calculations give us this table of payoffs for the "Desperation" strategy:

P_1	P_2	P_3	P_4
0.3537	0.3348	0.2410	0.070

Figure 1: Desperation Strategy Payoffs

We note that these values are fairly close to those given by the simulation. Now we need to compare them to the probabilities of the other strategies.

Probabilities of Victory: "Patience"

With the patience strategy, things are simpler.

$$P(P_1 \text{ duels } P_4) = \left(\frac{1}{4}\right)\left(\frac{1}{2}\right) = \frac{1}{8}$$
$$P(P_1 \text{ duels } P_3) = \left(\frac{3}{4}\right)\left(\frac{3}{7}\right) = \frac{9}{28}$$
$$P(P_1 \text{ duels } P_2) = \left(\frac{3}{4}\right)\left(\frac{4}{7}\right) + \left(\frac{1}{4}\right)\left(\frac{1}{2}\right) = \frac{12}{28} + \frac{1}{8} = \frac{31}{56}$$

In the end, we get the probability of player one winning via a wait strategy is:

$$\left(\frac{2}{5}\right)\left(\frac{31}{56}\right) + \left(\frac{4}{13}\right)\left(\frac{9}{28}\right) + \left(\frac{1}{4}\right)\left(\frac{1}{8}\right) = \frac{5119}{14560} \approx 0.3515$$

Some additional computations yield the remaining probabilities:

P_1	P_2	P_3	P_4
0.3515	0.3321	0.2225	0.0937

Figure 2: Patience Strategy Payoffs

Probabilities of Victory: "Revolution"

$$P(P_1 \text{ duels } P_4) = \left(\frac{3}{4}\right)^2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{4}\right) = \frac{9}{256}$$
$$P(P_1 \text{ duels } P_3) = \frac{3}{32}\sum_{n=0}^{\infty}\left(\frac{3}{32}\right)^n + \frac{9}{32}\sum_{n=0}^{\infty}\left(\frac{3}{32}\right)^n + \left(\frac{9}{32}\right)^2\sum_{n=0}^{\infty}\left(\frac{3}{32}\right)^n = \frac{465}{928}$$

Now, because P_1 does not necessarily go first when left alone with P_2 in this strategy, we need to compute the probability P_1 is left alone with P_2 and the probability that P_1 beats P_2 together. The relevant calculation is tedious and is thus relegated to the appendix. For now, just know that it comes out to be $\frac{363}{2080}$. So the probability of P_1 winning under the "Revolution Strategy" is

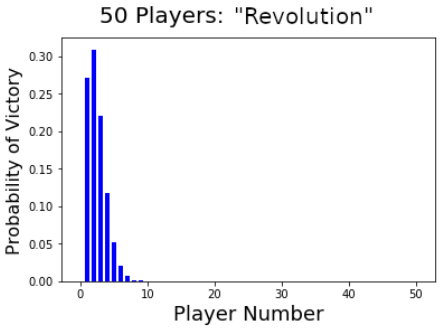
$$\frac{9}{256}\left(\frac{1}{4}\right) + \frac{465}{928}\left(\frac{4}{13}\right) + \frac{363}{2080} = \frac{651429}{1230240} \approx 0.3374$$

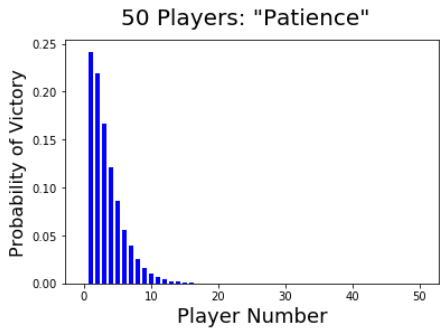
Remark. I have reason to believe this value is erroneous and should be closer to 0.321583, but I'm not sure where the arithmetic error lies.

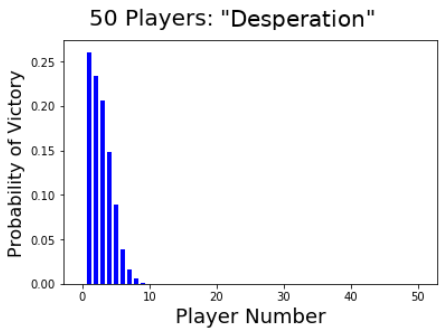
We note that no player can improve their chances of winning by deviating from the "Desperation" configuration, so it is a pure strategy Nash Equilibrium by definition.

Non-cooperative duels for n players

The four player case suggests that the first few players possess an enormous advantage over the last several players. It is not particularly surprising that this trend persists and becomes more extreme as the number of players increases. On the next page, there is a plot of the simulated probabilities of a given player winning under a given strategy in a 50 player game.







It would appear as if "Desperation" remains a Nash Equilibrium for the 50-player case, though we note that it is no longer the ideal outcome for the first player (and second). If the approximate distributions pictured above are any indication, at some point a "Revolution" strategy becomes more optimal for the first player. However, we know that it is not a Nash Equilibrium because the third from last player in a given cycle will still always choose to intentionally miss (and survive the current cycle with probability 1). In these games, we have been implicitly assuming that no collusion has been occurring. It may be optimal for the top players to cooperate and eliminate lower players, since lower players have an enormous advantage in these sequential duel games when no cooperation occurs. For the moment, we can conjecture the following if our space of strategies is restricted to the three we have highlighted thus far:

Conjecture. *In an n-person sequential duel where shooter s_i hits their target with probability $\frac{i}{n}$, "Desperation" is a pure-strategy Nash equilibrium.*

Appendix A

Revolution Simulation (Python)

```
import random
import numpy

numberPlayers = 50

survivals = numpy.zeros(numberPlayers)

accuracy = numpy.zeros(numberPlayers) # list of accuracy values
for i in range(0,(numberPlayers-1)):
    accuracy[i] = (i+1)/numberPlayers
accuracy[numberPlayers-1] = 1

def play_round():
        players = list(enumerate(accuracy))
        current = players[0]
        while len(players) > 1:
                targetPlayer = players[-1]
                if current == targetPlayer:
                        targetPlayer = players[-2]
                        #if self is targeted, switch to penultimate player

                good_shot = random.uniform(0,1)<=current[1]

                if good_shot:
                        players.remove(targetPlayer)
        # the counter moves on to the next standing player
                nextIndex = (players.index(current)+1)%(len(players))
                current = players[nextIndex]
        lastPlayer = players[0][0]
        survivals[lastPlayer] += 1
round_number = 10000
for _ in range(round_number):
        play_round()
print("Revolution", [x/round_number for x in survivals])
```

Patience Simulation (Python)

```
import random
import numpy

numberPlayers = 50

survivals=numpy.zeros(numberPlayers)

accuracy = numpy.zeros(numberPlayers)
for i in range(0,(numberPlayers-1)):
    accuracy[i]=(i+1)/numberPlayers
accuracy[numberPlayers-1] = 1

def play_round():
    players = list(enumerate(accuracy))
    current = players[0]
    while len(players) > 1:
        targetPlayer = players[-1]
        if current[0] < len(players)-2:
             nextIndex = (players.index(current)+1)%(len(players))
             current = players[nextIndex]


                #last living player is target
        else:
            if current == targetPlayer:
                targetPlayer = players[-2]
            good_shot = random.uniform(0,1)<=current[1]
            if good_shot:
                players.remove(targetPlayer)
            nextIndex = (players.index(current)+1)%(len(players))
            current = players[nextIndex]
    lastPlayer = players[0][0]
    survivals[lastPlayer] += 1
round_number = 100000
for _ in range(round_number):
        play_round()
print("Patience Strategy", [x/round_number for x in survivals])
```

Desperation Simulation (Python)

```
import random
import numpy
numberPlayers = 50
survivals = numpy.zeros(numberPlayers)
accuracy = numpy.zeros(numberPlayers)
for i in range(0,(numberPlayers-1)):
    accuracy[i] = (i+1)/numberPlayers
accuracy[numberPlayers-1]=1
def play_round():
    players = list(enumerate(accuracy))
    current = players[0]
    while len(players) > 1:
        targetPlayer = players[-1]
        if current[0] <= len(players)-4:
            if len(players) > 3:
                good_shot = random.uniform(0,1)<=current[1]
                if good_shot:
                    players.remove(targetPlayer)
            nextIndex = (players.index(current)+1)%(len(players))
            current = players[nextIndex]
        elif current[0] == len(players)-3:
            nextIndex = (players.index(current)+1)%(len(players))
            current = players[nextIndex]
        else:
            if current == targetPlayer:
                targetPlayer = players[-2]
            good_shot = random.uniform(0,1) <= current[1]
            if good_shot:
                players.remove(targetPlayer)
            #the counter moves on to the next player who is still alive.
            nextIndex = (players.index(current)+1)%(len(players))
            current = players[nextIndex]
    lastPlayer = players[0][0]
    survivals[lastPlayer] += 1

round_number=100000
for _ in range(round_number):
        play_round()

print("Desperation", [x/round_number for x in survivals])
```

Appendix B

We compute the probability that P_1 wins by beating P_2 in a duel as follows:

$$
\begin{aligned}
&\left(\frac{1}{4}\right)\left(\frac{1}{2}\sum_{n=0}^{\infty}\frac{3}{32}^{n}\right)\left(\frac{2}{5}\right) && P(P_1 \text{☠} P_4) * P(P_2 \text{☠} P_3) * P(P_1 \text{☠} P_2) \\
+&\left(\frac{1}{4}\right)\left(\frac{1}{2}\right)\left(\frac{1}{4}\right)\left(\frac{1}{4}\sum_{n=0}^{\infty}\frac{3}{32}^{n}\right)\left(\frac{1}{5}\right) && P(P_1 \text{☠} P_4) * P(P_1 \text{☠} P_3) * P(P_1 \text{☠} P_2) \\
+&\left(\frac{3}{4}\right)\left(\frac{1}{2}\right)\left(\frac{1}{4}\right)\left(\frac{3}{8}\sum_{n=0}^{\infty}\frac{3}{32}^{n}\right)\left(\frac{2}{5}\right) && P(P_1 \circ P_4) * P(P_2 \text{☠} P_4) * P(P_2 \text{☠} P_3) * P(P_1 \text{☠} P_2) \\
+&\left(\frac{3}{4}\right)\left(\frac{1}{2}\right)\left(\frac{1}{4}\right)\left(\frac{1}{4}\sum_{n=0}^{\infty}\frac{3}{32}^{n}\right)\left(\frac{1}{5}\right) && P(P_1 \circ P_4) * P(P_2 \text{☠} P_4) * P(P_1 \text{☠} P_3) * P(P_1 \text{☠} P_2) \\
+&\left(\frac{3}{4}\right)\left(\frac{1}{2}\right)\left(\frac{3}{4}\right)\left(\frac{3}{8}\sum_{n=0}^{\infty}\frac{3}{32}^{n}\right)\left(\frac{2}{5}\right) && P(P_1 \circ P_4) * P(P_2 \circ P_4) * P(P_3 \text{☠} P_4) * P(P_2 \text{☠} P_3) * P(P_1 \text{☠} P_2) \\
+&\left(\frac{3}{4}\right)\left(\frac{1}{2}\right)\left(\frac{3}{4}\right)\left(\frac{1}{4}\sum_{n=0}^{\infty}\frac{3}{32}^{n}\right)\left(\frac{1}{5}\right) && P(P_1 \circ P_4) * P(P_2 \circ P_4) * P(P_3 \text{☠} P_4) * P(P_1 \text{☠} P_3) * P(P_1 \text{☠} P_2) \\
+&\left(\frac{3}{4}\right)\left(\frac{1}{2}\right)\left(\frac{1}{4}\right)\left(\frac{1}{4}\right)\left(\frac{1}{5}\right) && P(\text{Everyone misses } P_4) * P(P_4 \text{☠} P_3) * P(P_1 \text{☠} P_4) * P(P_1 \text{☠} P_2) \\
+&\left(\frac{3}{4}\right)\left(\frac{1}{2}\right)\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)\left(\frac{1}{2}\right)\left(\frac{2}{5}\right) && P(\text{Everyone misses } P_4) * P(P_4 \text{☠} P_3) * P(P_2 \text{☠} P_4) * P(P_1 \text{☠} P_2)
\end{aligned}
$$

$$
= \left(\frac{32}{26}\right)\left(\frac{1}{20}+\frac{1}{640}+\frac{9}{640}+\frac{3}{640}+\frac{27}{640}+\frac{9}{640}\right)+\frac{3}{640}+\frac{9}{640}=\frac{363}{2080}
$$